

Batch: DMW 2 Roll No.: 16010324044

Experiment/assignment/tutorial No.: 05

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

TITLE: Decision Tree for classification and regression

AIM: a) To implement Decision Tree algorithm for classification and regression

b) To perform post pruning using minimal cost complexity pruning to prevent overfitting

OUTCOME: Students will be able to implement decision tree algorithm for classification and regression tasks and perform pruning to minimize overfitting:-

Procedure: Carefully note default parameters of sklearn function DecisionTreeClassifier and DecisionTreeRegressor separately, by visiting the respective websites

I) Classification

1. Read data, Heart.csv, and identify predictor and response variables; Perform encoding of categorical variables
2. Separate data into predictors and response variable; Split dataset into train and test sets and note size of each. Note down whether data is balanced or imbalanced data by counting data instances of each class. This will help to decide performance evaluation metric.
3. Use sklearn library function sklearn.tree.DecisionTreeClassifier to build full decision tree model with train data; Note down plotted tree and maximum depth of the tree
4. Perform prediction on train and test data and measure classification accuracy and other evaluation metrics. Compare train and test data accuracy and determine if pruning is required to prevent overfitting
5. Perform post pruning using minimal cost complexity pruning; Build tree models using effective values of alpha; Plot effective values of cost complexity parameter alpha vs total impurity of leaves; Plot number of nodes in a tree vs effective values of cost complexity parameter alpha; Plot depth of the tree vs effective values of cost complexity parameter alpha
6. Perform cross validation to find effective value of alpha based on average cross validation accuracy and determine value of alpha to prune the tree
7. Build pruned tree using chosen value of alpha; note down depth of the pruned tree
8. Perform prediction on train and test sets with trained pruned tree; note down accuracy and comment on effect of pruning on overfitting, if any

II) Regression

9. Reads data, SeoulBikeData.csv, and identify predictor and response variables; Perform one hot encoding of categorical variables
10. Split data into predictors and response variable; Split dataset into train and test sets and note size of each
11. Use sklearn library function `sklearn.tree.DecisionTreeRegressor` to build full decision tree model with train data; Note down plotted tree and maximum depth of the tree
12. Perform prediction on train and test data and measure Mean Square Error (MSE). Compare train and test data MSE and determine if pruning is required to prevent overfitting
13. Perform post pruning using minimal cost complexity pruning; Build tree models using effective values of alpha; Plot effective values of cost complexity parameter alpha vs total impurity of leaves; Plot number of nodes in a tree vs effective values of cost complexity parameter alpha; Plot depth of the tree vs effective values of cost complexity parameter alpha
14. Perform cross validation to find effective value of alpha based on average cross validation accuracy and determine value of alpha to prune the tree
15. Build pruned tree using chosen value of alpha; note down depth of the pruned tree
16. Perform prediction on train and test sets with trained pruned tree; note down MSE and comment on effect of pruning on overfitting, if any
17. Implement classification and regression on Raisin dataset and employee_data dataset respectively, using Decision Tree algorithm along with pruning and report the performance. Write a short note based on observations.

III] Miscellaneous: Play with few parameters of `DecisionTreeClassifier` & `DecisionTreeRegressor` functions and observe and note down their effect on performance

Observation Table: I) Classification

Observation No.	Observation / Analysis Question	Remark
1	Identify the predictor variables and response variable . Also mention the number of training and testing samples after train-test split.	The predictor variables provide the information required by the Decision Tree to classify whether the observation belongs to a particular heart-disease class. The training data is used to build the model, while the testing data is used to evaluate its generalization performance.

Observation No.	Observation / Analysis Question	Remark
	The predictor variables are all the input/features of the Heart.csv dataset used to predict the outcome. The response variable is the target/heart-disease class variable. The dataset was divided into 80% training data and 20% testing data using the train-test split. Predictor variables: All features except the response variable Response variable: Target/heart-disease class Training samples: 688 Testing samples: 230	
2	What is the maximum depth of the full decision tree? What does the depth indicate about the complexity of the tree? The maximum depth of the full Decision Tree is 14 Tree depth represents the maximum number of decision levels from the root node to a leaf node. A greater depth indicates a more complex tree with more decision rules.	A very deep tree can model complicated relationships in the training data but may also learn noise and cause overfitting. A shallower tree is simpler and generally easier to interpret.
3	Compare the training accuracy and testing accuracy of the full decision tree. Is there any indication of overfitting? Training accuracy: 100 % Testing accuracy: 77 % If the training accuracy is considerably higher than the testing accuracy, the full Decision Tree shows an indication of overfitting.	A large difference between training and testing accuracy means that the model performs very well on data it has already seen but does not generalize equally well to unseen data. Therefore, pruning is required to reduce the complexity of the tree.

Observation No.	Observation / Analysis Question	Remark
4	From the confusion matrix/classification metrics, which class is better classified and which class has more misclassifications? From the confusion matrix and classification report: Better classified class: _____ Class with more misclassifications: _____ The class having higher recall/F1-score and fewer false predictions is considered better classified.	The confusion matrix shows how correctly the model identifies each class. A larger number of off-diagonal elements for a particular class indicates more misclassifications for that class. Fill these two class names only after checking your actual confusion matrix.
5	From the α vs total leaf impurity, α vs number of nodes, and α vs tree depth graphs, what happens to the tree as α increases? As the cost-complexity parameter α increases, the penalty for tree complexity increases. Therefore, more branches are removed from the tree. α increases $\rightarrow$ number of nodes decreases α increases $\rightarrow$ tree depth decreases α increases $\rightarrow$ tree becomes simpler Total leaf impurity generally increases because the tree becomes less detailed.	Increasing α produces stronger pruning. A high α results in a smaller and simpler tree, which can reduce overfitting but excessive pruning may lead to underfitting.
6	What value of α is selected using cross-validation? Why is this value preferred for pruning? The value of α selected using 5-fold cross-validation is: $\alpha = 0.19$ This value is preferred because it gives the best average cross-validation accuracy among the evaluated α values.	Cross-validation helps select a pruning parameter that provides a good balance between model complexity and classification performance. Therefore, the selected α is used to construct the final pruned tree.

Observation No.	Observation / Analysis Question	Remark									
7	Compare the full tree and pruned tree in terms of depth and test accuracy. Does pruning reduce overfitting? Justify your answer. <table> <tr> <td>Parameter</td><td>Full Tree</td><td>Pruned Tree</td></tr> <tr> <td>Depth</td><td>14</td><td>5</td></tr> <tr> <td>Test Accuracy</td><td>_____ %</td><td>_____ %</td></tr> </table> The pruned tree has a lower depth than the full tree, indicating reduced model complexity.	Parameter	Full Tree	Pruned Tree	Depth	14	5	Test Accuracy	_____ %	_____ %	If the testing accuracy of the pruned tree is higher or approximately equal to that of the full tree while its depth is lower, pruning successfully reduces overfitting and improves generalization. If testing accuracy decreases slightly, pruning has still simplified the model, but the reduction in complexity has come at the cost of some predictive performance.
Parameter	Full Tree	Pruned Tree									
Depth	14	5									
Test Accuracy	_____ %	_____ %									

II) Regression

Observation No.	Observation / Analysis Question	Remark
1	Identify the predictor variables and response variable. Also mention the number of training and testing samples after train-test split. Predictor variables: All except Rented Bike Count (and Date) Response variable: Rented Bike Count Training samples: 3000 Testing samples: 1000	The predictor variables are used by the Decision Tree Regressor to estimate the number of rented bikes. The training set is used for model construction and the testing set is used to evaluate prediction performance on unseen data.
2	What is the maximum depth of the full regression tree? What does this indicate about the model complexity? 26	The depth represents the number of successive decision levels in the regression tree. A high depth indicates a complex model with many decision rules. Although this allows the model to capture detailed patterns, an excessively deep tree may fit the training data too closely and result in overfitting.
3	Compare the training MSE and testing MSE of the full tree. Is there evidence of overfitting? Training MSE: 0.0	MSE measures the average squared difference between actual and predicted values. A large difference between training and testing MSE indicates that the model performs much better on the training data than on unseen data. This

Observation No.	Observation / Analysis Question	Remark
	Testing MSE: 15.04	suggests that pruning may improve generalization.
4	From the α vs total leaf impurity, α vs number of nodes, and α vs tree depth graphs, what happens to the tree as α increases? As α increases, the complexity penalty becomes stronger and branches with insufficient improvement are removed. Number of nodes decreases. Tree depth decreases. The tree becomes simpler. Total leaf impurity generally increases.	Increasing α results in stronger pruning. This reduces model complexity and can help prevent overfitting. However, an excessively large α can make the tree too simple and cause underfitting.
5	What value of α is selected using cross-validation? Why was this value selected? The α value selected using cross-validation is: $\alpha = \underline{\hspace{2cm}}$ It was selected because it produced the lowest average cross-validation MSE among the tested α values.	For regression, lower MSE indicates better prediction performance. Cross-validation therefore helps identify an α that provides good predictive performance while controlling the complexity of the regression tree.
6	Compare the full and pruned trees using test MSE. Has the prediction performance improved after pruning? Parameter Full Tree Pruned Tree Test MSE <u> </u> <u> </u> If the pruned tree has a lower test MSE, prediction performance improved after pruning. If the test MSE is approximately unchanged, pruning simplified the model without	The test MSE is the important measure for comparing generalization performance because it represents prediction error on unseen data.

Observation No.	Observation / Analysis Question	Remark
	significantly affecting prediction performance. If the test MSE increased, pruning reduced complexity but resulted in some loss of predictive accuracy.	
7	Based on the change in tree depth and test MSE , explain whether pruning was beneficial for the regression model. The full tree had a depth of _____, while the pruned tree had a depth of _____. The test MSE changed from _____ for the full tree to _____ for the pruned tree.	Pruning was beneficial if the tree depth decreased while the test MSE decreased or remained approximately the same. This indicates that pruning reduced unnecessary complexity and improved or maintained the model's ability to generalize. If the test MSE increased substantially after pruning, then pruning was not beneficial from a prediction-accuracy perspective, although it did produce a simpler model.

Conclusion: -

In this experiment, the Decision Tree Classifier successfully classified the heart-disease data. The full tree was more complex and may show overfitting when training accuracy is substantially higher than testing accuracy. MCCPruning reduced the depth and number of nodes of the tree. The optimal α was selected using cross-validation to obtain a suitable balance between model complexity and classification accuracy.

Simi in this experiment, the Decision Tree Regressor was used to predict bike rental counts. The difference between training and testing MSE was used to identify possible overfitting. Use MCCPruning to reduce depth and find balance between model complexity and classification accuracy just as above.

Signature of faculty in-charge with date